

SMART INDIA HACKATHON 2026

PROJECT REPORT

Smart Competency Development Platform with iGOT Integration

Problem Statement ID: SIH26101

Problem Statement: Smart Competency Development Platform with iGOT
Integration

Theme: Smart Education **Category:** Software

Team ID: 136984

Team Name: Recursion

AI-Enabled Personalized Learning and Competency Development Platform

Academic Project Report

CERTIFICATE

This is to certify that the project report entitled “**Smart Competency Development Platform with iGOT Integration**” has been prepared as an academic project based on the Smart India Hackathon 2026 problem statement SIH26101. The work describes an AI-enabled learning platform for competency assessment, personalized learning, learning-material processing, and quiz/MCQ generation, with the intended integration of the iGOT Karmayogi ecosystem.

Project Team	Recursion
Team ID	136984
Problem Statement ID	SIH26101

Project Guide / Faculty Signature: _____

Date: _____ Place: _____

DECLARATION

We declare that this project report is prepared for academic and project-submission purposes and presents the proposed Smart Competency Development Platform with iGOT Integration. The report is organized around the official SIH 2026 problem-statement template supplied for this project. Technical details that are described as proposed technologies or future integrations should be treated as implementation plans unless demonstrated by the prototype.

Team Name: Recursion

Team ID: 136984

ACKNOWLEDGEMENT

We express our sincere gratitude to our faculty members, mentors, institution, and the Smart India Hackathon platform for providing an opportunity to work on a real-world competency-development problem. We also acknowledge the public resources and documentation of iGOT Karmayogi, Karmayogi Bharat, MoSPI, Django, Python, and Google AI that support the research and technical direction of this project.

ABSTRACT

The Smart Competency Development Platform with iGOT Integration is proposed as an AI-enabled learning solution for identifying competency gaps and supporting personalized learning. The platform is designed to assess learner skills, identify areas requiring improvement, recommend relevant training, process uploaded learning materials, and generate quizzes and multiple-choice questions (MCQs). The system uses a web-based architecture centered on Python and Django, with HTML, CSS, JavaScript and Bootstrap for the interface. AI/LLM capabilities are proposed for competency-gap analysis and question generation, while PDF-processing support can be provided through PyMuPDF. The project is aligned with the Smart Education theme of Smart India Hackathon 2026 and the supplied problem statement SIH26101.

Keywords: iGOT Karmayogi, competency development, personalized learning, AI, LLM, quiz generation, MCQ generation, Django, Smart Education.

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1. INTRODUCTION

The digital transformation of learning systems creates an opportunity to move from generic training toward competency-oriented and personalized learning. The supplied SIH 2026 template identifies the project as a Smart Competency Development Platform with iGOT Integration under the Smart Education theme and software category.

The proposed platform focuses on a learner-centered cycle: assessment, competency-gap identification, personalized training recommendation, learning-material processing, AI-assisted quiz generation, assessment, and progress tracking. The objective is to make learning more targeted and measurable rather than relying only on static course consumption.

The platform is intended to support the broader competency-development goals associated with the iGOT Karmayogi ecosystem. Integration points are therefore treated as an important architectural requirement, while the exact production API/data-access mechanism depends on the interfaces made available by the ecosystem.

2. PROBLEM STATEMENT AND MOTIVATION

The official project template supplied for this report names the problem statement **SIH26101 – Smart Competency Development Platform with iGOT Integration**.

A conventional learning portal can present courses and tests, but it may not automatically connect learner performance to specific competency gaps and personalized next steps. Learners may also have large quantities of learning material without an efficient mechanism to convert that material into practice questions.

The project therefore proposes a single workflow that connects competency assessment with recommendations and AI-assisted assessment content. This provides a practical bridge between learning resources, learner performance, and continuous skill development.

3. OBJECTIVES AND SCOPE

Objectives

- Identify competency gaps from learner assessment and performance data.
- Recommend personalized learning or training resources based on identified gaps.
- Process uploaded learning materials for learning support.
- Generate quizzes and MCQs from relevant learning content.
- Track assessment results and learner progress.
- Provide a web-based, accessible learning interface.
- Provide an architecture capable of connecting with iGOT Karmayogi data/services where authorized interfaces are available.

Scope

The initial scope is a web-based prototype. It covers learner registration, skill assessment, gap detection, course recommendation, material upload, AI processing, quiz generation, assessment, and progress tracking. Advanced enterprise-scale deployment, production identity integration, official API certification, large-scale analytics, and high-availability infrastructure are treated as future or deployment-stage concerns unless separately implemented.

4. LITERATURE REVIEW AND RESEARCH

The project research references listed in the supplied template include iGOT Karmayogi, Karmayogi Bharat, MoSPI, Django, Python, and Google AI.

iGOT Karmayogi: The platform is considered the ecosystem context for competency-oriented learning and training integration. The proposed system is designed to complement such an ecosystem through personalized learning and competency analytics.

AI in Education: AI/LLM methods can support content transformation, question generation, and learner-support workflows. In this project, AI is positioned as an assistance layer rather than a replacement for learning governance.

Adaptive Learning: The competency-gap concept follows an adaptive-learning direction: learner evidence is used to determine what should be practiced next.

Web Application Engineering: Django and Python provide the proposed server-side foundation, while standard web technologies provide the user interface.

5. PROPOSED SYSTEM

The proposed system combines a learner portal, assessment engine, competency-analysis layer, recommendation layer, learning-material processor, AI question-generation service, and progress dashboard.

Core flow: User Registration → Skill Assessment → Gap Detection → Course Recommendation → Material Upload → AI Processing → Quiz Generation → Assessment → Progress Tracking.

Innovation: Instead of treating courses, documents, and tests as independent components, the platform connects them into a competency-development loop. A learner can be assessed, receive targeted recommendations, study material, practice generated questions, and use the resulting performance as new evidence for progress tracking.

6. SYSTEM ARCHITECTURE AND WORKFLOW

High-level architecture

The browser-based client communicates with a Django application server. The application manages authentication, learner data, assessments, recommendations, uploaded materials, quiz records, and progress information. An AI/LLM service is connected to the application for competency-gap analysis and content/question generation. A database stores structured application data. A document-processing layer extracts useful text from supported PDF materials. An integration layer is reserved for authorized iGOT Karmayogi services or available data sources.

Workflow

1. Learner registers/logs in.
2. Learner completes a skill/competency assessment.
3. Assessment results are analyzed to identify weak areas.
4. Relevant learning resources are recommended.
5. Learner uploads permitted learning material when required.
6. PDF/document text is extracted and prepared for AI processing.
7. AI generates quizzes/MCQs from the processed content.
8. Learner attempts the assessment.
9. Results are stored and used for progress tracking.
10. The next learning cycle can use updated competency evidence.

Stage	Input	Output
Assessment	Learner responses	Scores
Gap Analysis	Scores + competency model	Gap areas
Recommendation	Gap areas	Learning resources
AI Quiz	Learning material	MCQs/Quiz
Tracking	Attempt results	Progress indicators

7. TECHNICAL APPROACH

Programming Languages: Python and JavaScript.

Framework: Django.

Frontend: HTML, CSS, JavaScript and Bootstrap.

AI/ML: Generative AI / LLM for competency-gap analysis and quiz generation.

Database: SQLite / MySQL depending on deployment needs.

Document Processing: PyMuPDF – PDF processing.

Integration: iGOT Karmayogi API / available authorized data sources.

Hardware: Laptop/PC with internet connectivity; optional cloud server for deployment.

The supplied SIH template explicitly reserves a Technical Approach page and links to the working prototype, report, GitHub repository, and diagram.

Development methodology: requirement analysis → UI and architecture design → database design → core module implementation → AI integration → testing → prototype deployment → documentation.

8. FUNCTIONAL MODULES

- 1. User Registration and Authentication:** Provides account creation and controlled access to learner features.
- 2. Skill Assessment:** Captures competency evidence through structured assessments.
- 3. Competency Gap Detection:** Analyzes assessment outcomes to identify areas needing improvement.
- 4. Personalized Recommendation:** Maps identified gaps to relevant learning resources.
- 5. Learning Material Upload:** Allows supported materials to be supplied for learning and question generation.
- 6. AI Quiz/MCQ Generator:** Produces practice questions from processed learning content.
- 7. Assessment Engine:** Presents questions, captures responses, and calculates results.
- 8. Progress Tracking:** Maintains learner performance and improvement history.
- 9. Dashboard:** Presents relevant learning and performance information in a consolidated interface.

9. DATABASE AND DATA FLOW

The proposed database stores entities such as users, competencies, assessments, questions, learning materials, recommendations, quiz attempts, and progress records. A relational model is appropriate for maintaining consistent links between users, assessments, resources, and results.

Example entities: User(User_ID, Name, Email, Role), Competency(Competency_ID, Name, Description), Assessment(Assessment_ID, User_ID, Date, Score), Material(Material_ID, User_ID, File_Name, Extracted_Text), Question(Question_ID, Material_ID, Question_Text, Options, Answer), Attempt(Attempt_ID, User_ID, Quiz_ID, Score), Recommendation(Recommendation_ID, User_ID, Resource_ID, Reason).

Data flow: User data and assessment responses enter the application → analysis produces competency evidence → recommendations are generated → learning material is processed → questions are generated → quiz results are recorded → progress indicators are updated.

10. IMPLEMENTATION METHODOLOGY

Phase 1 – Requirement Analysis: Understand the SIH problem statement, target users, learning workflow, and required outputs.

Phase 2 – UI/UX Design: Create a simple learner dashboard and module navigation for assessment, learning, quizzes, and progress.

Phase 3 – Backend Development: Implement Django models, views, URLs, authentication, assessment logic, and APIs.

Phase 4 – AI Layer: Connect an appropriate LLM service and define controlled prompts for competency analysis and question generation.

Phase 5 – Document Processing: Extract text from supported PDFs using PyMuPDF and prepare clean text for downstream processing.

Phase 6 – Testing: Test authentication, assessments, question generation, scoring, storage, and UI flows.

Phase 7 – Deployment: Package the application for local or cloud deployment and document the integration requirements.

11. FEASIBILITY AND VIABILITY

The supplied template contains a dedicated Feasibility and Viability section.

Technical Feasibility: The core system can be built using widely used web technologies such as Python, Django, JavaScript, relational databases, PDF processing, and an AI service.

Operational Feasibility: The learner workflow is based on familiar actions—login, assessment, learning, quiz, and progress review—so the interface can remain straightforward.

Economic Feasibility: A prototype can be developed with open-source software and modest computing resources. AI/API and cloud costs depend on usage and deployment scale.

Integration Feasibility: The architecture can expose an integration layer, but production integration depends on official iGOT interfaces, authorization, data formats, security requirements, and access policies.

Viability: The concept can evolve from a prototype into a broader competency-support platform through stronger analytics, validated recommendations, secure integrations, and scalable infrastructure.

12. CHALLENGES, RISKS AND MITIGATION

AI-generated inaccuracies: Generated questions may contain factual or contextual errors. Mitigation: validation rules, source-grounding, human review for high-stakes content, and quality sampling.

Data privacy: Learner and organizational data may be sensitive. Mitigation: access control, secure storage, minimal data collection, encryption, logging, and policy-aligned retention.

Integration dependency: Official iGOT APIs/data sources may have access or technical constraints. Mitigation: modular integration adapters and clear separation between core learning functions and external services.

Document quality: Scanned or poorly structured PDFs can reduce extraction quality. Mitigation: preprocessing, OCR where permitted, validation, and user feedback.

Scalability: AI processing and concurrent learners can increase infrastructure demand. Mitigation: asynchronous processing, caching, queue-based jobs, and cloud scaling where needed.

13. IMPACT AND BENEFITS

The supplied SIH template includes a dedicated Impact and Benefits section.

Educational: Learners receive more targeted practice and can identify weak competency areas.

Organizational: Training can be connected more directly with competency evidence and learner progress.

Accessibility: A web-based interface can make learning resources available across devices with internet access.

Productivity: Automated generation of practice questions can reduce the manual effort required to create basic assessment material.

Scalability: The modular architecture can support additional competencies, content sources, assessment types, and analytics over time.

14. TESTING AND EVALUATION PLAN

Functional testing: Verify registration, login, assessment submission, scoring, material upload, quiz generation, and progress tracking.

AI quality evaluation: Check question relevance, correctness, uniqueness, answer validity, difficulty alignment, and source consistency using sampled test sets.

Performance testing: Measure response time for common application actions and processing time for uploaded materials.

Security testing: Check authentication, authorization, upload validation, input sanitization, and protection of stored data.

User acceptance testing: Collect feedback on navigation, clarity, usefulness of recommendations, and ease of completing quizzes.

15. FUTURE SCOPE

Future development can include official iGOT integration after access and API requirements are established; multilingual learning; richer competency graphs; role-based dashboards; advanced learning analytics; recommendation personalization using longitudinal performance; voice-based learning support; stronger document understanding; automated difficulty calibration; and cloud-native scaling.

The system can also be extended to support administrators and training managers with aggregate, privacy-aware insights, competency trends, content effectiveness measures, and training-planning tools.

16. CONCLUSION

The Smart Competency Development Platform with iGOT Integration presents a structured approach to competency-oriented digital learning. By connecting assessment, gap identification, personalized recommendations, learning-material processing, AI-generated quizzes, and progress tracking, the proposed system creates a continuous learning loop.

The project is aligned with the supplied SIH 2026 problem statement SIH26101 and the Smart Education theme. The architecture is designed to remain modular so that the learning platform can operate as a prototype while providing a clear path toward authorized iGOT ecosystem integration.

17. REFERENCES

1. iGOT Karmayogi — <https://igotkarmayogi.gov.in/>
2. Karmayogi Bharat — <https://karmayogibharat.gov.in/>
3. Ministry of Statistics and Programme Implementation (MoSPI) — <https://www.mospi.gov.in/>
4. Django Documentation / Website — <https://www.djangoproject.com/>
5. Python — <https://www.python.org/>
6. Google AI — <https://ai.google/>
7. Research literature on adaptive learning and intelligent tutoring systems.
8. Documentation and technical references for PyMuPDF and database technologies used during implementation.

18. APPENDIX: PROJECT LINKS

Project resources:

The supplied SIH template is designed to expose links for the Project Report, Live Prototype, GitHub Repository, and Diagram.

GitHub: <https://github.com/shubh-jaiswal-897/MENTAL-ABILITY-TEST-SOFTWARE>

Prototype: <https://mental-ability-test-softwares.vercel.app/>

Report: To be inserted with the final hosted report URL.

Note: The above GitHub and prototype addresses are included from the project information provided for this report; the uploaded SIH template itself only establishes link placeholders for these resources.